Sammathamu Vandana Sahani

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.80
Tirumala Junior College	Class XII (AP State board)	2023	97.80%
Dr. K. K. R's Gowtham School	Class X (AP State board)	2021	99.66%

WORK EXPERIENCE

Technology Analyst Intern

Modernised the existing **UI design** of Gemini-COIN page

Developed entire UI for COIN module without solely using standard MSUI components

Implemented **light theme UI** for the entire COIN module

PROJECTS

Undergrad Researcher

Implemented a **parallelized approximate Betweenness Centrality algorithm**

Designed and compared **four source-sampling strategies**

Benchmarked the implementation on **real-world social network datasets** (Twitter and Google+)

Built an end-to-end **analysis pipeline**

Notification Framework

Built a full-stack **multi-channel notification platform**

Implemented **secure user authentication**

Added **rate-limiting** and client management features

Developed webhook and **media-handling** capabilities

Multi-client Quiz System

Built a **multithreaded TCP client-server application** in C++

Implemented a thread-safe, **real-time leaderboard** system

Designed a custom **application-layer protocol**

Validated protocol correctness using **packet-level network analysis**

Vehicle Rental System

Implemented **role-based authentication** and access control

Built a complete vehicle **inventory management** module

Developed a staff management dashboard

Designed a **booking and user management system** for staff/admin use

RISC-V Cache Simulator

Built a **RISC-V assembler and functional simulator** from scratch in C++

Designed and integrated a configurable **cache simulator**

Implemented a **cycle-accurate instruction execution** engine

Validated correctness through **hand-written RISC-V assembly test programs**

SKILLS

Languages: C++, C, RISC-V

Tech Framework:

Frontend: JavaScript, HTML, CSS, TypeScript, Angular basics

Backend & Databases: SQL, Node.js, MongoDB, FastAPI, django basics

Development Tools: Linux, Git, Docker, Jupyter, MySQL Workbench

RELEVANT COURSES

Computer Science: Software Engineering, Data Structres and Algorithms , Computer Architecture, Computer Networks, Compilers, Operating Systems, Database Management Systems, Deep learning, Foundations of Machine Learning

Mathematics: Probability and Random Variables,Intro Linear Algebra, Transform Techniques, Calculus

POSITIONS OF RESPONSIBILITY

Web Development Core, Sunshine Team IITH 2024-2025

Design Core, Alumni Cell IITH 2025-2026

SCHOLASTIC ACHIEVEMENTS

Got C.V. Raman '**Young Genius**' award for scoring top rank in National/State level Science Talent Search Exam 2014

Secured **school 1st** in 29th National Mathematics Talent Search (Ramanujan) 2018

EXTRACURRICULAR

Awarded **Star Performer** for my tenure as web developer **Sunshine IITH**, 2024-2025

Interests: Sketching, painting, and handcrafted art

Volunteer, NSS [2023-2027] - assisted in planning and coordinating events like Vidhyadaan, Half marathon.